caseSetup-v4.1.1 Technical User Guide

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This document covers the use of caseSetup for external automotive aerodynamics cases.

Contents

1 Introduction

caseSetup was built over a couple years for the Ontario Tech Racing team. This process is a mix of experiences gained from FSAE related activities and professional activities. The first iteration of caseSetup was born in the fall of 2022. The drive behind caseSetup was to improve the consistency of runs from user to user while reducing the chances of mistakes. The main idea behind it was to use pre-defined templates which mildly restricts the user from making un-intended modifications to the setup. This allows for consistent meshing, solving, and post processing regardless of user.

This document here is the technical user guide, while some Computational Fluid Dynamics (CFD) theories may be included, it is not meant to be an exhaustive explination or discussion of the physics. Please refer to specialized CFD books and resources if more information is required.

2 Directory Structure

This process requires a specific directory structure for your cases to work. It looks like the following:

- JOB-DIRECTORY
  - 01_incoming(YYYY-MM-DD)
    - 2025-09-11_New_Geometry
  - 02_references
    - ANSA
      - trial009.ansa.gz
    - MSH
      - trial009.obj.gz
      - fw-v1-1.obj.gz
  - 03_reports
    - 051_064_063_report_2025-11-16.pptx
  - CASES
    - 001
    - 002
    - 003
    - ...
    - 009
    -
      - **caseSetup**
      - constant
      - system

The definitions for each folder is described here:

- 01_incoming(YYYY-MM-DD) - Contains all the incoming geometry from other departments or other files which are required for the simulation. Each instance of CAD delivered to the department must be in a dated folder with a short descriptive name.
- 02_references - Contains all the geometry files that are used in the simulation. Subdirectories are ANSA and MSH
- ANSA - Contains all the ANSA files that are used in the simulation, ideally contains the CAD files within them and not just the tri-surfaces. Should be saved as ".gz" for smaller file size.

- MSH - Contains all the tri-surfaces that are used in the simulation. Ideally this is where the original copies of all the STLs or OBJ files live. Each trial folder contains only a symbolic link referencing to this directory to reduce the number and size of geometry files.
- 03_reports - Contains all the reports placed by the pptGeneration process or placed by the user manually.
- CASES - Contains all the simulation runs, this is where each case/trial folder lives, each trial is denoted by a 3 digit number XXX.
- caseSetup - This is the file that controls the simulations and the outputs.
- system - This directory contains the files required for the simulation to run.
- constant - Like system this directory also contains the files required for the simulation and also in constant/triSurfaces is where the geometry for the simulation is stored for this case. It is ideally a symbolic link so that repeated geometries are not copied many times in many cases.

Once all these directories are setup, we are ready to explore the rest of the features.

3 Basic Setup

To start, you would need a case directory created in your CASES directory. Once created, you will not have a caseSetup file just yet. To get a file, execute the following line in your case directory,

```
caseSetup --new
```

A fresh caseSetup file will be generated. The inside of the file will look like this:

```
[DEFAULTS]
DEFAULT_MODULES = defaultGlobalControl defaultGlobalSimControl defaultBCSetup defaultGlobalDecomposition
     defaultGlobalMaterial defaultGlobalRefinement defaultPost
ADDON_MODULES =

[TITLES]
CASENAME = casename
CASEDESCRP = default description
JOBCODE = TS

[GEOMETRY]
GEOM = default.stl,1,8,5,1.1,high
```

The DEFAULT_MODULES line controls the settings that are available. If the setting is not visible, it just means that it will use the default setup. Each setting here is explained below:

- CASENAME - the case name for your case, it does not do anything at this time, but in the future it could be added to the post processing images.
- CASEDESCRP - case description, see CASENAME
- JOBCODE - Two letter initial for the job you are running, just helps identify the job that's running on the cluster queue.
- GEOM - List of geometries to use in the case. A comma separates each setting for that particular geometry, the next geometry goes on the subsequent line.
  - What each column represents:

    ```
    GEOMETRY_FILE_NAME,SCALE,MESH_LEVEL,INFLATION_LAYERS,INFLATION_LAYER_GROWTH_RATE,
        WALL_MODEL
    ```

Each geometry that is required in this run and is in its own file should be listed here, the settings shown above will be applied to all geometries in that file. If a different settings are required to be applied, that geometry should be broken out into it's own file. A common example is wheels.

3.1 Rotating Geometries

The wheels and tires are spinning and therefore should have a rotating wall boundary condition applied, therefore it MUST be in its own part. Example,

```
[DEFAULTS]
DEFAULT_MODULES = defaultGlobalControl defaultGlobalSimControl defaultBCSetup defaultGlobalDecomposition
     defaultGlobalMaterial defaultGlobalRefinement defaultPost
ADDON_MODULES =

[TITLES]
CASENAME = casename
CASEDESCRP = default description
JOBCODE = TS

[GEOMETRY]
GEOM = body.obj.gz,1,8,5,1.1,high
        ROTA-fr-wh-lhs,1,8,5,1.1,high
        ROTA-fr-wh-rhs,1,8,5,1.1,high
        ROTA-rr-wh-lhs,1,8,5,1.1,high
        ROTA-rr-wh-rhs,1,8,5,1.1,high
```

Notice the prefix ROTA, it lets caseSetup know that this geometry will have a rotating wall boundary condition applied. Upon running caseSetup, if it's the first time with this set of geometry, caseSetup will stop and let you know that the following blocks have been added. The user should check that these are the appropriate settings for each wheel geometry.

NOTE: If using a plinth, ensure that the plinth is a separate geometry, else the automatic wheel center calculation will include the plinth in the radius calculation resulting in an incorrect radius and angular velocity!

```
[ROTA-fr-wh-lhs]
WH_RAD = default
WH_CENTER = default
WH_AXIS = default
ROT_WH = True

[ROTA-fr-wh-rhs]
WH_RAD = default
WH_CENTER = default
WH_AXIS = default
ROT_WH = True

[ROTA-rr-wh-lhs]
WH_RAD = default
WH_CENTER = default
WH_AXIS = default
ROT_WH = True

[ROTA-rr-wh-rhs]
WH_RAD = default
WH_CENTER = default
WH_AXIS = default
ROT_WH = True
```

- `WH_RAD` [[`float,m`]] - wheel radius, distance from wheel center to ground (`r`, `default`)
- `WH_CENTER` [[`float,m`]] [[`float,m`]] [[`float,m`]] - wheel center coordinates in global coordinates system (`x y z`, `default`)
- `WH_AXIS` [[`float,m`]] [[`float,m`]] [[`float,m`]] - wheel axis of rotation from, using right hand rule with respect to the global coordinate system (`x y z`, `default`)
- `WH_AXIS` [`bool`] - activate rotating wall or use no-slip condition (`true`, `false`)

Using `default` option will trigger the automatic calculation but manual input of values is still possible. To speed up `caseSetup` it is recommended to copy the values output by the calculation in the termial and input them into the caseSetup if coordinates and axis values are not expected to change for subsequent runs.

3.2 Controlling Run Settings

Deleting the module `defaultGlobalControl` will expose the following block:

```
[GLOBAL_CONTROL]
STARTFROM = latestTime
STARTTIME = 0
ENDTIME = 1000
DT = 1
WRITEINT = 500
PURGEINT = 1
AVGSTART = 500
```

- `STARTFROM` [`str`] - which time to start the simulation at, `latestTime` will start the simulation at the lastest available time if running for an existing solution, `startTime` will have the simulation start at the value specified in `STARTTIME` (`latestTime`, `startTime`)
- `STARTTIME` [`float/int,s`] - time to start the simulation at, will be used if `STARTFROM` is specified as `startTime`, must use `int` if steady run is used, or `float` if a transient run is selected
- `ENDTIME` [`float/int,s`] - final time or time step to end the simulation at, must use `int` if steady run is used, or `float` if a transient run is selected
- `DT` [`float/int,s`] - timestep size, must use `1` if steady run is used, or any `float` if a transient run is selected

- WRITEINT [float/int,s] - interval which the data is written to disk for restart or post-processing use, must use int if steady run is used, or any float if a transient run is selected (**CAUTION:** saving too frequently can cause increased solve time and increased storage usage!)
- PURGEINT [int] - number of written timesteps to keep, will only keep PURGEINT number of time steps (aside from STARTTIME or 0)
- DT [float/int,s] - when to start averaging flow variables, must use int if steady run is used, or float if a transient run is selected

3.3 Controlling Simulation Settings

Deleting the module defaultGlobalSimControl will expose the following block:

```
[GLOBAL_SIM_CONTROL]
SIM_TYPE = steady
SIM_INIT = potential
SIM_SYM = half
TURB_MODEL = kosst
INIT_P = default
INIT_K = default
INIT_OMEGA = default
INIT_NUT = default
INIT_NUTILDA = default
```

- SIM_TYPE [str] - what type of simulation (steady, transient)
- SIM_INIT [str] - what type of initialization, steady only available when SIM_TYPE = transient (none, potential, steady)
- SIM_SYM [str] - simulation symmetry type, reccomended to run in half if not doing yaw and vehicle is mostly symmetrical, use full if running internal flow (half, full)
- TURB_MODEL [str] - turbulence model applied, transient runs can only use sa ([kosst,sa])
  - steady
    - kosst - $k - \omega$ Shear Stress Transport ($k - \omega$ SST)
    - sa - Spalart-Allmaras
  - transient
    - sa - Spalart-Allmaras Delayed Detached Eddy Simulation (SADDES)
- INIT_P [float,Pa] - initial conditions for pressure, reccomended to leave in default unless specific values are required (float, default)
- INIT_K [float,m2s-2] - initial conditions for turublent kinetic energy, reccomended to leave in default unless specific values are required (float, default)
- INIT_OMEGA [float,s-1] - initial conditions for ω term, reccomended to leave in default unless specific values are required (float, default)
- INIT_NUT [float,m2s-1] - initial conditions for ν_t term, reccomended to leave in default unless specific values are required (float, default)
- INIT_NUTILDA [float,m2s-1] - initial conditions for $\tilde{\nu}$ term, reccomended to leave in default unless specific values are required (float, default)

3.4 Boundary Conditions

Deleting the module defaultBCSetup will expose the following block:

```
[BC_SETUP]
INLET_MAG = 15
YAW = 0
DRAG_VEC = default
LIFT_VEC = default
PITCH_VEC = default
GROUND = moving
REFCOR = 0.775 0 0
REFLEN = 1.55
REFWIDTH = 1
REFAREA = 1
POROSITY = yes
BASE_CELL_SIZE = 1.6
DOMAIN_SIZE = 48 16 16
DOMAIN_OFFSET = 0.5
FRT_WH_CTR = 0 0 0
```

- INLET_MAG [float,ms-1] - inlet velocity magnitude ([float])
- YAW [float,deg] - yaw angle, will only yaw in the positive direction, if SIM_SYM is half, this value not used (float)
- DRAG_VEC [[float,m] [float,m] [float,m]] - drag force vector, following SAE J1594 convention by default (x y z, default)
- LIFT_VEC [[float,m] [float,m] [float,m]] - lift force vector, following SAE J1594 convention by default (x y z, default)
- PITCH_VEC [[float,m] [float,m] [float,m]] - pitch moment vector, following SAE J1594 convention by default (x y z, default)
- GROUND [str] - ground plane boundary condition (moving,stationary)
- REFCOR [[float,m] [float,m] [float,m]] - point defining center of rotation, typically center of wheel projected to the ground (x y z)
- REFLEN [[float,m]] - reference length, typically wheelbase length (float)
- REFWIDTH [[float,m]] - reference width, typically trackwidth length (float)
- REFAREA [[float,m]] - reference area, typically frontal area or 1 to have calculations done in $C_D A$ (float)
- POROSITY [str] - to calculate forces including forces due to porous media, recommend to leave at yes (yes, no)
- BASE_CELL_SIZE [[float,m]] - base cell size of blockMesh, resulting mesh levels are calculated based off this number, recommend to leave at 1.6 (float)
- DOMAIN_SIZE [[float,m] [float,m] [float,m]] - dimensions for blockMesh domain (x y z)
- DOMAIN_OFFSET [[float,m]] - distance of FRT_WH_CTR to the inlet boundary as a ratio of domain x length, i.e. 0.5 means the FRT_WH_CTR is located at the center of the domain (float)
- FRT_WH_CTR [[float,m] [float,m] [float,m]] - coordinate for the center point projected to the ground of the front wheel axle (x y z)

3.5 Decomposition and Cluster Control

Deleting the module defaultGlobalDecomposition will expose the following block:

```
[GLOBAL_COMPUTE_SETUP]
QUEUE_TYPE = slurm
SCRIPT_VERSION = 1
NUM_NODES = 1
TASK_PER_NODE = 72
DECOMP = 18 2 2
```

- QUEUE_TYPE [str] - NOT ACTIVE, FOR FUTURE USE (str)
- SCRIPT_VERSION [int] - NOT ACTIVE, FOR FUTURE USE (int)
- NUM_NODES [int] - NOT ACTIVE, FOR FUTURE USE (int)
- TASK_PER_NODE [int] - number of cores to use (int)
- DECOMP [int int int] - decomposition pattern, dictates the number of divisions in each direction during decomposition, freestream direction should be the largest, the product of the 3 values must match TASK_PER_NODE (x y z)

3.6 Fluid Properties

Deleting the module defaultMaterial will expose the following block:

```
[GLOBAL_MATERIAL]
NPHASE = 1
MATERIAL_TYPES = air
VISCOSITY = 1.46e-5
```

- NPHASE [int] - NOT ACTIVE, FOR FUTURE USE (int)
- MATERIAL_TYPES [str] - NOT ACTIVE, FOR FUTURE USE (str)
- VISCOSITY [float, m2s-1] - fluid kinematic viscosity (float)

3.7 Meshing Control

Deleting the module defaultGlobalRefinement will expose the following block:

```
[GLOBAL_REFINEMENT]
TEMPLATE_TYPE = snappyTemplate1
MIN_GEOM_LEVEL = 8
REF_FEAT_ANGLE = 25
GEOM_REF_DIST = 0.01 0.05 0.1
GEOM_REF_LEVELS = 9 8 7
DEFAULT_WAKE_REF = defaultWake
LOC_IN_MESH = -1 -1 4
LAYER_TYPE = absolute
DEF_EX_RATIO = 1.2
REF_ANGLE = 25
FEAT_EDGE_ANGLE = 140
FEAT_EDGE_LEVEL_INC = 1
FIRST_LAYER_HEIGHT = 0.0001
FINAL_LAYER_HEIGHT = 0.001
FINAL_LAYER_RATIO = 0.3
```

- TEMPLATE_TYPE [str] - meshing template to use (str)
  - snappyHexMesh - template file must exist in setupTemplates/<caseSetup_template>/defaultDicts/
  - ansaMesh - TEMPLATE_TYPE string must include caseSetup template name as well as ansa separated by underscores i.e. otr_ansa_default. The ansa is used to trigger the use of ansaMesh. The template must be located in setupTemplates/<caseSetup_template>/defaultANSA/.
    The last value entry in TEMPLATE_TYPE is the name of the template within that path and can be of any length. Just NO SPACES!

- MIN_GEOM_LEVEL [int] - coarses mesh level to be used for all geometries in the case (int)
- REF_FEAT_ANGLE [float,deg] - angle where curvatures with angles larger than it would trigger mesh refinement in that area, operates only for ansaMesh (float)
- GEOM_REF_DIST [[float,m] [float,m] [float,m] ...] - distance from the surface which the mesh level from GEOM_REF_LEVELS shall be applied, unlimited entries, but must match length in GEOM_REF_LEVELS (float float ...)
- GEOM_REF_LEVELS [int int ...] - refinement levels for each corresponding distance in GEOM_REF_DIST unlimited entries, but must match length in GEOM_REF_DIST (int int ...)
- DEFAULT_WAKE_REF [str] - wake refinement generated by default or use specific template (true, false, str)
  - if using specific template, it should exist in setupTemplates/<caseSetup_template>/defaultRefinements/
- LOC_IN_MESH [[float,m] [float,m] [float,m]] - point which defines the "inside" location of the domain, only works for snappyHexMesh (float float)
- LAYER_TYPE [str] - how the layer growth occurs globally, only works for ansaMesh, defined in snappyHexMesh by template (absolute, aspect)
  - absolute - first layer height is defined absolutely based on FIRST_LAYER_HEIGHT, rest of inflation layer sizing determined by aspect ratio set for the geometry
  - aspect - first layer height is defined as a ratio to cell size outside of the inflation layers, global settings currently hardcoded as 0.5, rest of inflation layer sizing determined by aspect ratio set for the geometry
- DEF_EX_RATIO [float] - global layer expansion ratio (float)
- REF_ANGLE [float,deg] - angle where curvatures with angles larger than it would trigger mesh refinement in that area, operates only for snappyHexMesh (float)
- FEAT_EDGE_ANGLE [float,deg] - angle under which triggers edge refinement, only works for snappyHexMesh (float)
- FEAT_EDGE_LEVEL_INC [int] - refinement level increment for edges defined by FEAT_EDGE_ANGLE, only works for snappyHexMesh (int)
- FIRST_LAYER_HEIGHT [[float,m]] - first layer height, only works for ansaMesh (float)
- FINAL_LAYER_HEIGHT [[float,m]] - total height of inflation layers, not currently operational, for future use (float)
- FINAL_LAYER_RATIO [[float,m]] - ratio of last layer and the surrounding element size, not currently operational, for future use (float)

3.8 Post Processing Control

Deleting the module defaultPost will expose the following block:

```
[GLOBAL_POST_PRO]
ISOQ = 2000
ISOCPT = 0
SLICES = True
XSLICE = default
YSLICE = default
ZSLICE = default
```

- ISOQ [float,s-2] - Q criterion value of which to generate VTK iso surfaces of (float)

- ISOCPT [float] - $C_{p,total}$ value of which to generate VTK iso surfaces of (float)
- SLICES [str] - to generate VTK slices (true, false)
- XSLICE [[float,m] [float,m] [float,m]] - start, end and spacing of the x normal slices (float, float, float)
- YSLICE [[float,m] [float,m] [float,m]] - start, end and spacing of the y normal slices (float, float, float)
- ZSLICE [[float,m] [float,m] [float,m]] - start, end and spacing of the z normal slices (float, float, float)

3.9 Ride Height Control

Deleting the module defaultRideHeight will expose the following block that is under development currently. The goal is to create a process which can do ride height mapping.

```
[RIDE_HEIGHT_SETUP]
RUN_RIDE_HEIGHT = false
RIDE_HEIGHT_FILE =
USE_DEPEND = True
```

This section outlines the usage of the ride height mapping tool. A ride height map is useful to understand the dynamic sensitivity of an aero package as the car goes through the corners. We can use it to run in virtual lap time simulations as well. The use of the ride height mapping tool is diverse, we can use it for evaluation of a concept to understand the overall sensitivity from concept to concept. Some concepts perform better under braking, or under acceleration.

For the sake of simplicity, we make the assumption that there is enough time for the air flow to settle and there is no hysterisis from attitude to attitude on the track, although this might not be true. Therefore, each ride height is run statically with no movement in the simulation after transformation.

3.9.1 Ride Height Map Generation

Generating a ride height map is done by using on track or simulations data, looking at the front ride height vs rear ride height scatter plots and creating a point grid that encompases the data. There might be some outliers that rarely get reached, but it is helpful to add a point there to ensure the map is interpolated well. A full ride height map can be many points, which can result in very long total computational times. They should not be something run day to day. Rather a limit selection of ride heights can be run for conceptual analysis. The full map can be run at the end of a design cycle to evaluate the overall sensitivity.

For day to day usage, the ride height points can be limited to 3-5 points relevant to the geometry concept at hand. For example, a front wing change it might be more relevant to use select nose down and nose down plus roll conditions in addition to a level condition.

3.9.2 Ride Height Tool Operation

The ride height tool operates by reading a .csv file containing front left (fl), front right (fr), rear left (rl), and rear right (rr) ride heights as a deviation from the baseline condition. The ride height translation is done with a car centric coordinate system, meaning the car body does not move, only the wheels do as well as the domain. This gives the easiet comparison between ride heights in post processing. The ride height reference points are currently referenced to the axle location in x and z directions at the wheel base center in y. This is also assuming your front and rear wheelbases are the same. Future iterations will include alternate reference point locations.

Additionally, the model used for the ride height mapping should not contain suspension components as they will not be transformed at this time. Future iterations might include suspension kinematic transformations.

3.9.3 Ride Height Tool Usage

The use the ride height tool, delete the `defaultRideHeight` module in the `DEFAULT_MODULES` line in `caseSetup`. This will reveal the ride height module in `caseSetup`.

4 Prefix Controls

There are 3 or 4 letter prefixs that allow special control of your geometry and simulation. These help with assigning porous media regions, Multiple Reference Frame (MRF) regions, or even more control over meshing for certain geometries.

4.1 Rotating Geometries

The ROTA prefix allows the assignment of rotating wall to geometries such as wheels. A usage example is shown below.

```
[DEFAULTS]
DEFAULT_MODULES = defaultGlobalControl defaultGlobalSimControl defaultBCSetup defaultGlobalDecomposition
     defaultGlobalMaterial defaultGlobalRefinement defaultPost
ADDON_MODULES =

[TITLES]
CASENAME = casename
CASEDESCRP = default description
JOBCODE = TS

[GEOMETRY]
GEOM = body.obj.gz,1,8,5,1.1,high
        ROTA-fr-wh-lhs,1,8,5,1.1,high
        ROTA-fr-wh-rhs,1,8,5,1.1,high
        ROTA-rr-wh-lhs,1,8,5,1.1,high
        ROTA-rr-wh-rhs,1,8,5,1.1,high
```

Notice the prefix ROTA, it lets caseSetup know that this geometry will have a rotating wall boundary condition applied. Upon running caseSetup, if it's the first time with this set of geometry, caseSetup will stop and let you know that the following blocks have been added. The user should check that these are the appropriate settings for each wheel geometry.

NOTE: If using a plinth, ensure that the plinth is a separate geometry, else the automatic wheel center calculation will include the plinth in the radius calculation resulting in an incorrect radius and angular velocity!

```
[ROTA-fr-wh-lhs]
WH_RAD = default
WH_CENTER = default
WH_AXIS = default
ROT_WH = True

[ROTA-fr-wh-rhs]
WH_RAD = default
WH_CENTER = default
WH_AXIS = default
ROT_WH = True

[ROTA-rr-wh-lhs]
WH_RAD = default
WH_CENTER = default
WH_AXIS = default
ROT_WH = True

[ROTA-rr-wh-rhs]
WH_RAD = default
WH_CENTER = default
WH_AXIS = default
ROT_WH = True
```

- WH_RAD [[float,m]] - wheel radius, distance from wheel center to ground (r, default)
- WH_CENTER [[float,m]] [[float,m]] [[float,m]] - wheel center coordinates in global coordinates system (x y z, default)
- WH_AXIS [[float,m]] [[float,m]] [[float,m]] - wheel axis of rotation from, using right hand rule with respect to the global coordinate system (x y z, default)
- WH_AXIS [bool] - activate rotating wall or use no-slip condition (true, false)

Using default option will trigger the automatic calculation but manual input of values is still possible. To speed up caseSetup it is recommended to copy the values output by the calculation in the termial and input them into the caseSetup if coordinates and axis values are not expected to change for subsequent runs.

4.2 Porous Media

When radiators are used in a simulation, it would be very computationally expensive to model each radiator fin. So instead we use the Darcy-Forcheimer porous media calculation to model the pressure drop across a radiator which allows us to estimate the mass flow across the radiator and resulting forces.

The prefix `POR` is used to define the porous media geometry. **CAUTION:** The porous media geometry must only contain the "box" or region that represents the radiator fin region, do not include the frame or any other part of the radiator. The box which represents the fin region should be a simple prism. The inlet and outlets should have their own PID with the keyword `-inlet` or `-outlet`. There should only be 1 of each PID!

The mass flow calculations are performed on each pid in the porous media, therefore including the walls or any other surfaces where flow isn't moving *across* would make the calculations erronous.

Including the prefix `POR` on a geometry would expose the following block:

```
GEOM = trial005-rad.obj.gz,1,8,6,1.4,high
        fw-H05A00.obj.gz,1,8,6,1.4,high
        rw-R11I01.obj.gz,1,8,6,1.4,high
        POR-rad-001.obj.gz,1,10,6,1.4,high
        ROTA-fr-wh-lhs-v3-3.obj.gz,1,8,6,1.4,high
        ROTA-fr-wh-rhs-v3-3.obj.gz,1,8,6,1.4,high
        ROTA-rr-wh-lhs-v3-3.obj.gz,1,8,6,1.4,high
        ROTA-rr-wh-rhs-v3-3.obj.gz,1,8,6,1.4,high

[POR-rad-001]
DCOEFFS = 1000
FCOEFFS = 1000
VEC1 = 0.7071 0 -0.7071
VEC2 = 0 1 0
POINT = 1.7875 -0.000583 0.342739
SAMPLE_POR = True
```

the name of the block will be the name of the geometry just without the extension to identify it.

- `DCOEFFS` [`float`] - Darcy coefficient attained from experimental testing of pressure drop vs flow rate of the radiator (`float`)
- `FCOEFFS` [`float`] - Forcheimer coefficient attained from experimental testing of pressure drop vs flow rate of the radiator (`float`)
- `VEC1` [[`float,m`] [`float,m`] [`float,m`]] - vector defining the normal of the radiator exit (`float float float`)
- `VEC2` [[`float,m`] [`float,m`] [`float,m`]] - vector in plane with the radiator exit (`float float float`)
- `POINT` [[`float,m`] [`float,m`] [`float,m`]] - point defining the "inside" of the radiator (`float float float`)
- `SAMPLE_POR` [`str,m`] - to sample the mass flow through the radiator (`true`, `false`)